Budget: 100 kgCO_2 pro m Gurtlänge (inkl. Transport LKW)

Median

10%- / 90%-Fraktil

N_{Rd} [MN]

Zuggurt

Vollholz C24

Brettschichtholz GL24h

Baustahl S355

50 mm

100 mm

40 mm

14 mm

u [mm]

Stahlbeton C30/37, vorgespannt (10 MPa) ρ_p = 0.9%

100 mm

50 mm

32 mm

Betonstahl B500B (nackt)

Stahlbeton C30/37, ρ_s=0.6%

Umschnürter Beton C30/37 ρ_w=8.6%, ρ_l=0.6%

Druckgurt

Stahlbeton C30/37
Zug: ρ_s = 10%
A = 139'392 mm^2
g_{0k} = 4.1 kN/m
Druck: ρ_s = 0.6%
A = 420'833 mm^2
g_{0k} = 10.2 kN/m

Umschnürter Beton
ρ_w = 8.6%, ρ_l = 0.6%
A = 143'484 mm^2
g_{0k} = 4.5 kN/m

Brettschichtholz
GL24h
A = 625'000 mm^2
g_{0k} = 2.8 kN/m

Vollholz
C24
A = 552'486 mm^2
g_{0k} = 2.6 kN/m

vorgespannter Beton
σ_{cp} = 10 MPa
ρ_p = 0.9%
A_p = 3'057 mm^2
A_c = 338'330 mm^2
g_{0k} = 8.4 kN/m

Baustahl
A = 18'605 mm^2
g_{0k} = 1.4 kN/m

Betonstahl
A = 18'829 mm^2
g_{0k} = 1.4 kN/m